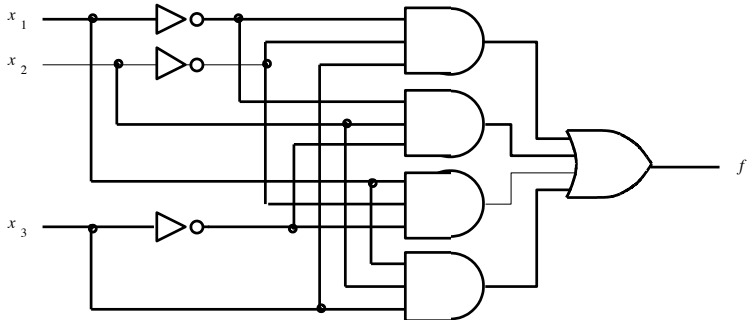
Department of Electrical and Computer Engineering
Queen's University

ELEC-271 Digital Systems

Solutions to Problem Set 2

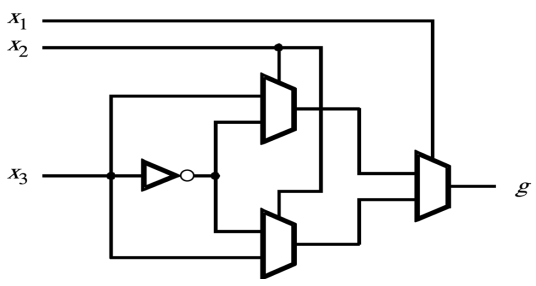
1. Consider the circuit shown below. Show the truth table for the logic function *f*. If each gate in the circuit is implemented as a CMOS gate, how many transistors are needed?



x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

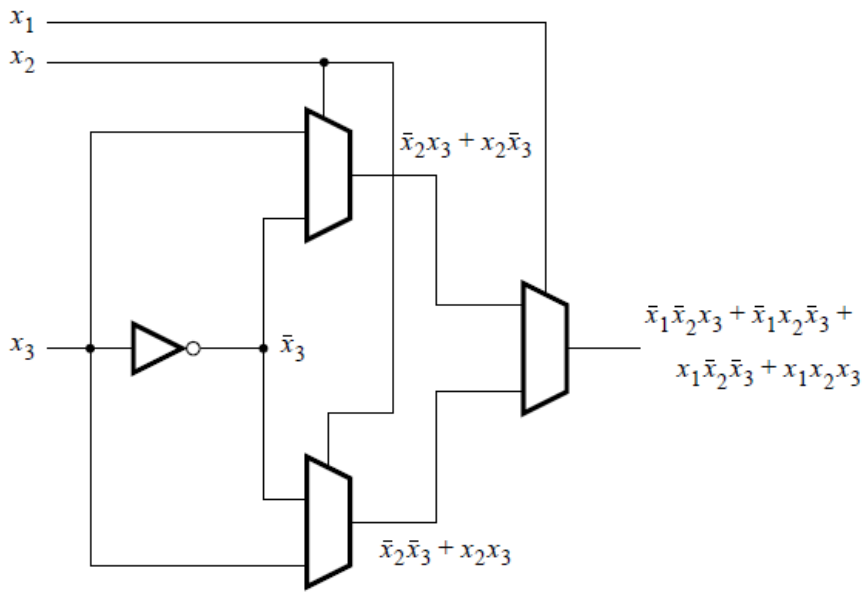
transistors = NOT gates x 2 + AND gates x 8 + OR gates x 10
= 3x2 + 4x8 + 1x10 = 48

2. Show that the following circuit is functionally equivalent to the circuit in question 1.

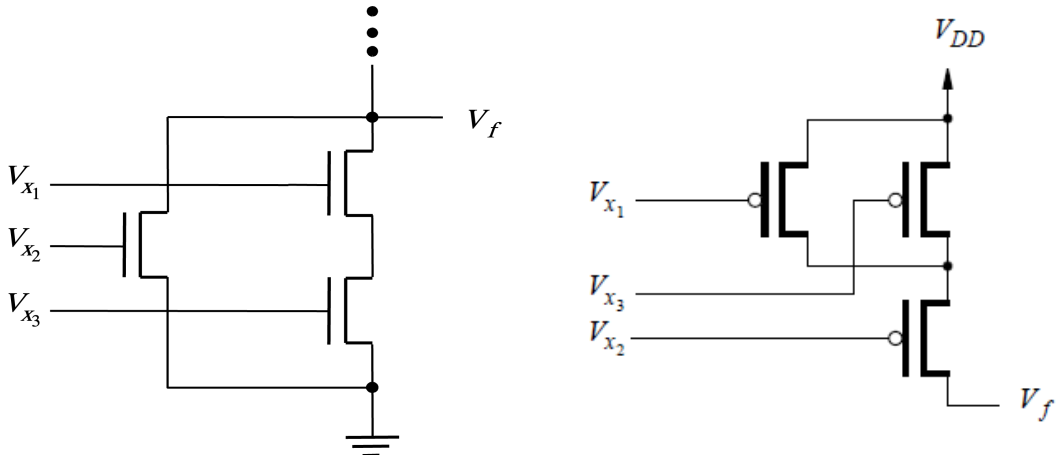


In question 1, the canonical SOP for f is:

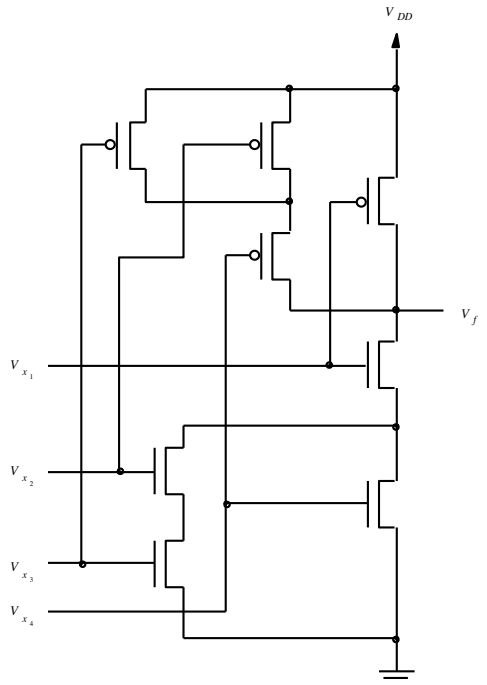
$$f = \overline{x}_1\overline{x}_2x_3 + \overline{x}_1x_2\overline{x}_3 + x_1\overline{x}_2\overline{x}_3 + x_1x_2x_3$$



3. The following circuit shows half of a CMOS circuit. Derive the other half that contains PMOS transistors.



4. Give the truth table for the CMOS circuit below. Derive a canonical SOP expression for the truth table.



The truth table:

x_1	x_2	x_3	x_4	f
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$$f(x_1,x_2,x_3,x_4) = \bar{x}_1\bar{x}_2\bar{x}_3\bar{x}_4 + \bar{x}_1\bar{x}_2\bar{x}_3x_4 + \bar{x}_1\bar{x}_2x_3\bar{x}_4 + \bar{x}_1\bar{x}_2x_3x_4 + \bar{x}_1x_2\bar{x}_3\bar{x}_4 + \bar{x}_1x_2\bar{x}_3x_4 + \bar{x}_1x_2x_3\bar{x}_4 + \bar{x}_1x_2x_3x_4 + x_1\bar{x}_2\bar{x}_3\bar{x}_4 + x_1\bar{x}_2\bar{x}_3x_4 + x_1\bar{x}_2x_3\bar{x}_4 + x_1x_2\bar{x}_3\bar{x}_4$$

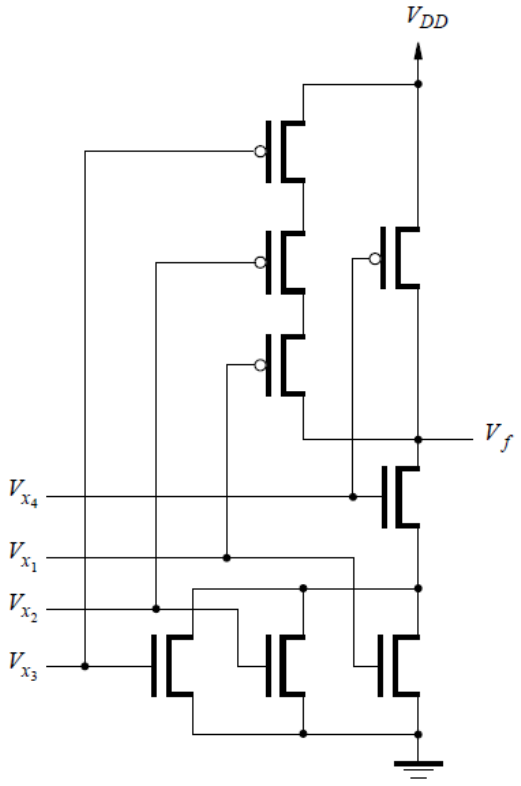
5. Derive a CMOS complex gate for the minimum SOP expression for the logic function $f(x_1,x_2,x_3,x_4) = \sum m(0,1,2,4,6,8,10,12,14)$.

$$f(x_1,x_2,x_3,x_4) = \bar{x}_1\bar{x}_2\bar{x}_3\bar{x}_4 + \bar{x}_1\bar{x}_2\bar{x}_3x_4 + \bar{x}_1\bar{x}_2x_3\bar{x}_4 + \bar{x}_1\bar{x}_2x_3x_4 + \bar{x}_1x_2\bar{x}_3\bar{x}_4 + \bar{x}_1x_2\bar{x}_3x_4 + \bar{x}_1x_2x_3\bar{x}_4 + x_1\bar{x}_2\bar{x}_3\bar{x}_4 + x_1\bar{x}_2\bar{x}_3x_4$$

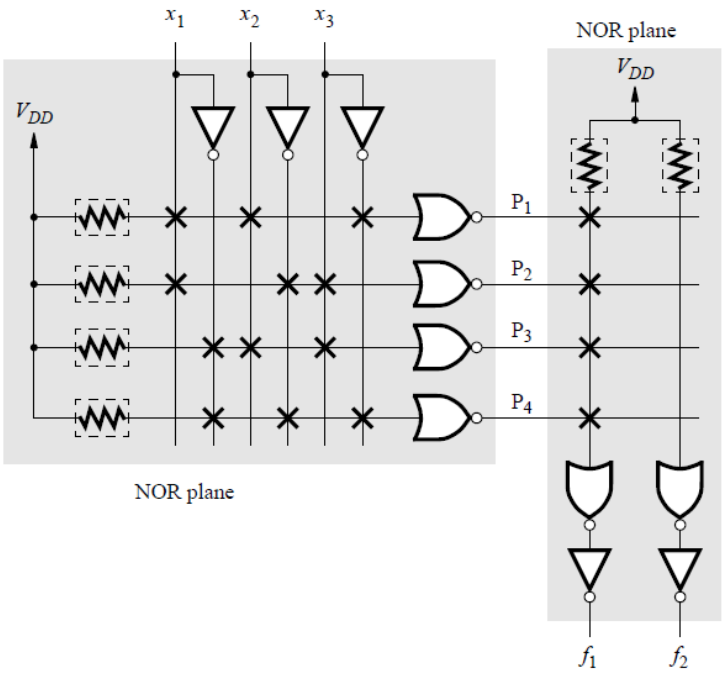
$$f(x_1,x_2,x_3,x_4) = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1x_3\bar{x}_4 + \bar{x}_1x_2\bar{x}_3\bar{x}_4 + x_1\bar{x}_2\bar{x}_4 + x_1\bar{x}_2x_4 + \bar{x}_1x_2x_3x_4 = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1x_3\bar{x}_4 + \bar{x}_1x_3x_4 + x_1\bar{x}_2\bar{x}_4 + x_1\bar{x}_2x_4$$

$$f(x_1,x_2,x_3,x_4) = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1x_3\bar{x}_4 + \bar{x}_1x_3x_4 + x_1\bar{x}_2\bar{x}_4 + x_1\bar{x}_2x_4 = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1\bar{x}_4 + x_1\bar{x}_4 = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_4$$

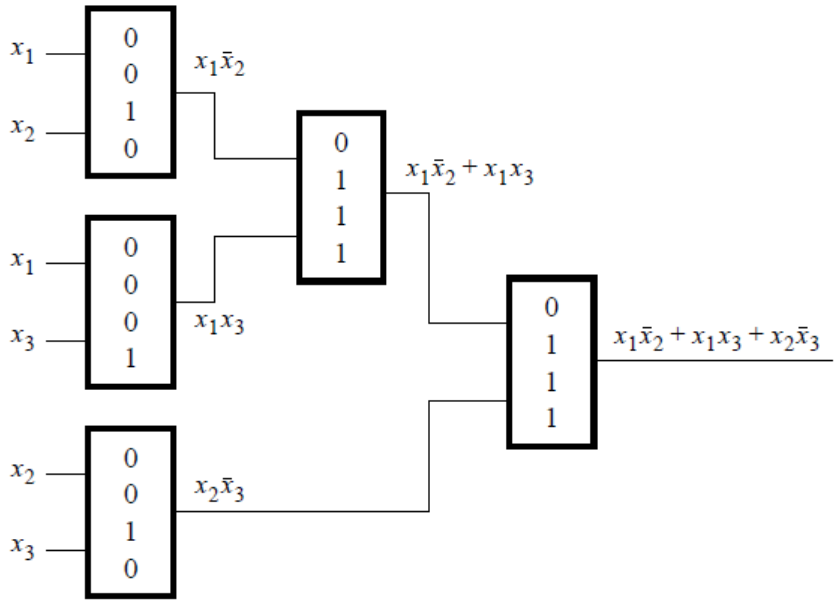
Minimum SOP expression for f is $f(x_1,x_2,x_3,x_4) = \bar{x}_4 + \bar{x}_1\bar{x}_2\bar{x}_3$



6. Using the style of drawing in Figure 3.66 of textbook, draw a picture of a PLA programmed to implement $f(x_1, x_2, x_3) = \sum m(1,2,4,7)$. The PLA should have the inputs x_1, x_2, and x_3; the product terms P_1, P_2, P_3, and P_4; and the outputs f_1 and f_2.



7. Consider the function $f(x_1, x_2, x_3) = x_1\bar{x}_2 + x_1x_3 + x_2\bar{x}_3$. Show a circuit using 5 two-input lookup-tables (LUTs) to implement this expression. Give the truth table implemented in each LUT. You do not need to show the wires in the FPGA.



8. Assume that a gate array contains the type of logic cell depicted in the following figure. The inputs in_1, in_2, …, in_7 can be connected to either 1, 0, or any logic signal. Show how the logic cell can be used to realize $f = x_1x_2 + x_3$.

